Reinforcement Learning

Alapan Kuila

Assistant Professor
Dept Of Computer Science and Engineering

Indian Institute of Information Technology Design and Manufacturing Kurnool

Acknowledgement

Slides and images used in this presentation are adapted from publicly available resources on Hugging Face (`https://huggingface.co/`) for educational purposes.

Types of Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

How Reinforcement Learning Differs

- In Reinforcement Learning, the agent does not know the environment beforehand.
- Instead, it learns(?) through interaction and rewards.
- The agent learns a *policy*.

Example

- state [Drone Position, sensor reading]
- Actions [move forward, turn left/right, change altitude]
- Reward[+100 for reaching goal, 100 for collision, 1 for energy use]

Reinforcement Learning: The big picture

- The idea behind Reinforcement Learning is that an agent will learn from the environment by interacting with it (through trial and error) and receiving rewards (negative or positive) as feedback for performing actions.
- Learning from interactions with the environment comes from our natural experiences.

The Reinforcement Learning Framework

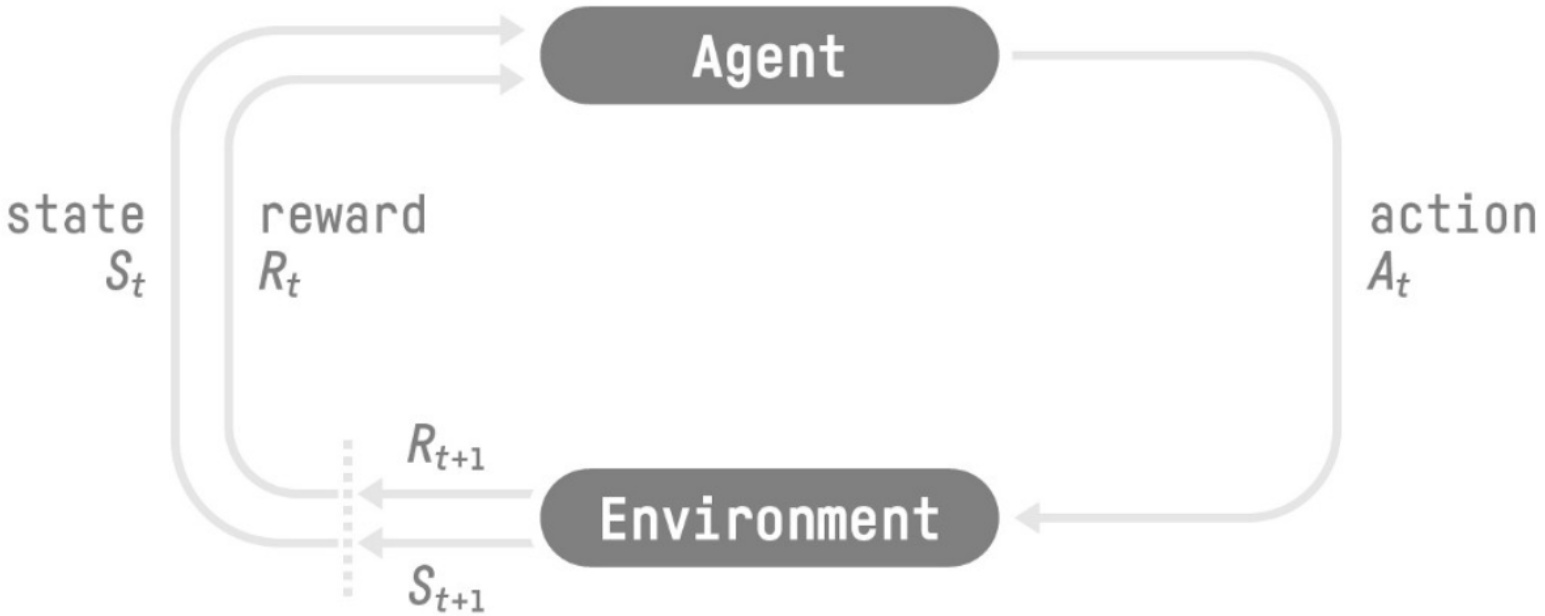


Figure: The RL Process: a loop of state, action, reward and next state

RL Process

This RL loop outputs a sequence of state, action, reward and next state.

RL Loop

- Our Agent receives state S_0 from the Environment
- Based on that state S_0, the Agent takes action A_0
- The environment goes to a new state S_1
- The environment gives some reward R_1 to the Agent



The Reward Hypothesis: the central idea of RL

The agent's goal is to maximize its cumulative reward, called the expected return.

Why is the goal of the agent to maximize the expected return?

Because RL is based on the reward hypothesis, which is that all goals can be described as the maximization of the expected return (expected cumulative reward).

That's why in Reinforcement Learning, to have the best behavior, we aim to learn to take actions that maximize the expected cumulative reward.

Markov Property

RL process is called a Markov Decision Process (MDP)

Note: RL agent needs only the current state to decide what action to take and not the history of all the states and actions they took before.

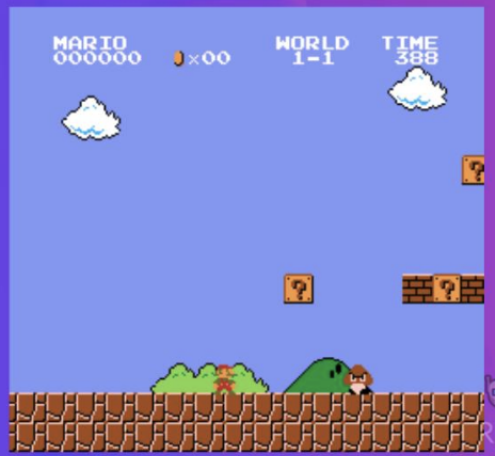
Observations/States Space

Observation Space

State: **complete description** of the state of the world (no hidden information).



Observation: **partial description** of the state of the world.



Action Space

Action Space

Discrete: **finite number of possible actions**



Continuous: **infinite number of possible actions**



Rewards and the discounting

The reward is fundamental in RL because it is the only feedback for the agent.

Thanks to it, our agent knows if the action taken was good or not.

$$R(\tau) = r_{t+1} + r_{t+2} + r_{t+3} + r_{t+4} + \dots$$

Return: cumulative reward

Trajectory (read Tau)
Sequence of states and actions

Figure: The cumulative reward equals the sum of all rewards in the sequence.

Rewards and the Discounting (contd.)

$$R(\tau) = \sum_{k=0}^{\infty} r_{t+k+1}$$

Figure: The cumulative reward equals the sum of all rewards in the sequence.

Rewards and the Discounting (contd.)

In reality, we can't just add the rewards like that! (Why?)

Rewards and the Discounting (contd.)

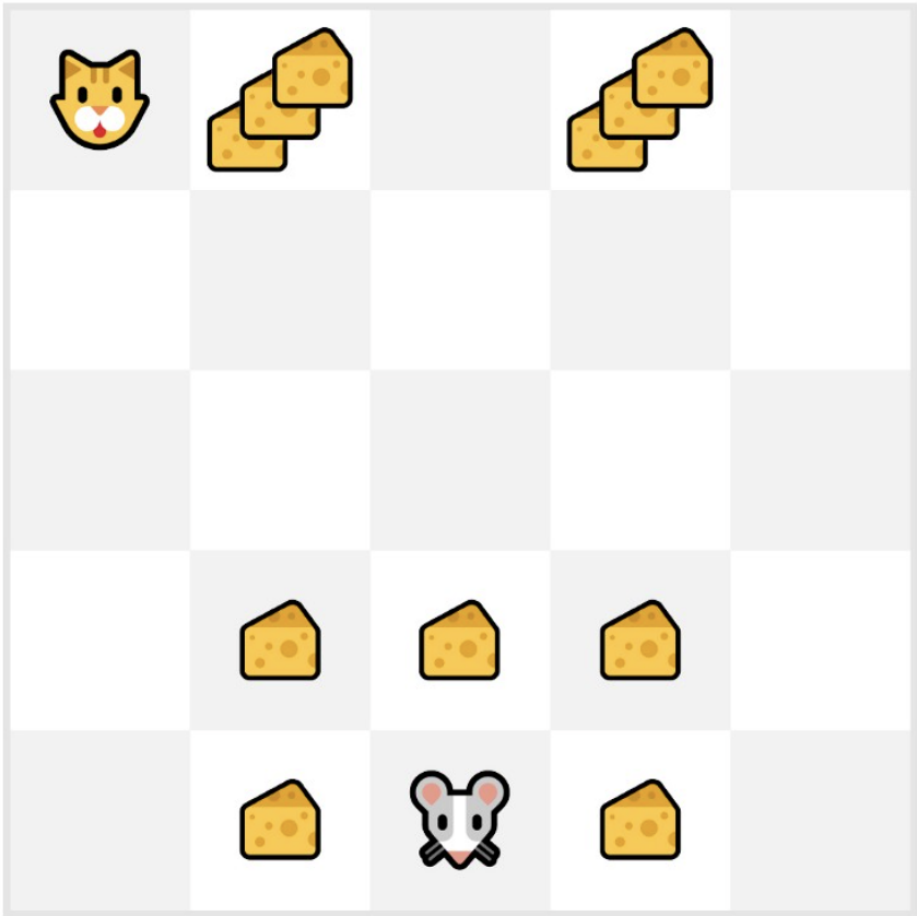


Figure: The reward near the cat, even if it is bigger, will be more discounted

Rewards and the Discounting (contd.)

$$R(\tau) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+4} + \dots$$

Return: cumulative reward

Gamma: discount rate

Trajectory (read Tau)
Sequence of states and actions

$$R(\tau) = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

Figure: The discounted expected cumulative reward function

Rewards and the Discounting (contd.)

To discount the rewards, we proceed like this:

- We define a discount rate called *gamma*. It must be between 0 and 1. Most of the time between 0.95 and 0.99.
  - The larger the *gamma*, the smaller the discount. This means our agent cares more about the long-term reward.
  - On the other hand, the smaller the *gamma*, the bigger the discount. This means our agent cares more about the short term reward.
- Then, each reward will be discounted by *gamma* to the exponent of the time step. As the time step increases, the cat gets closer to us, so the future reward is less and less likely to happen.

Type of Tasks

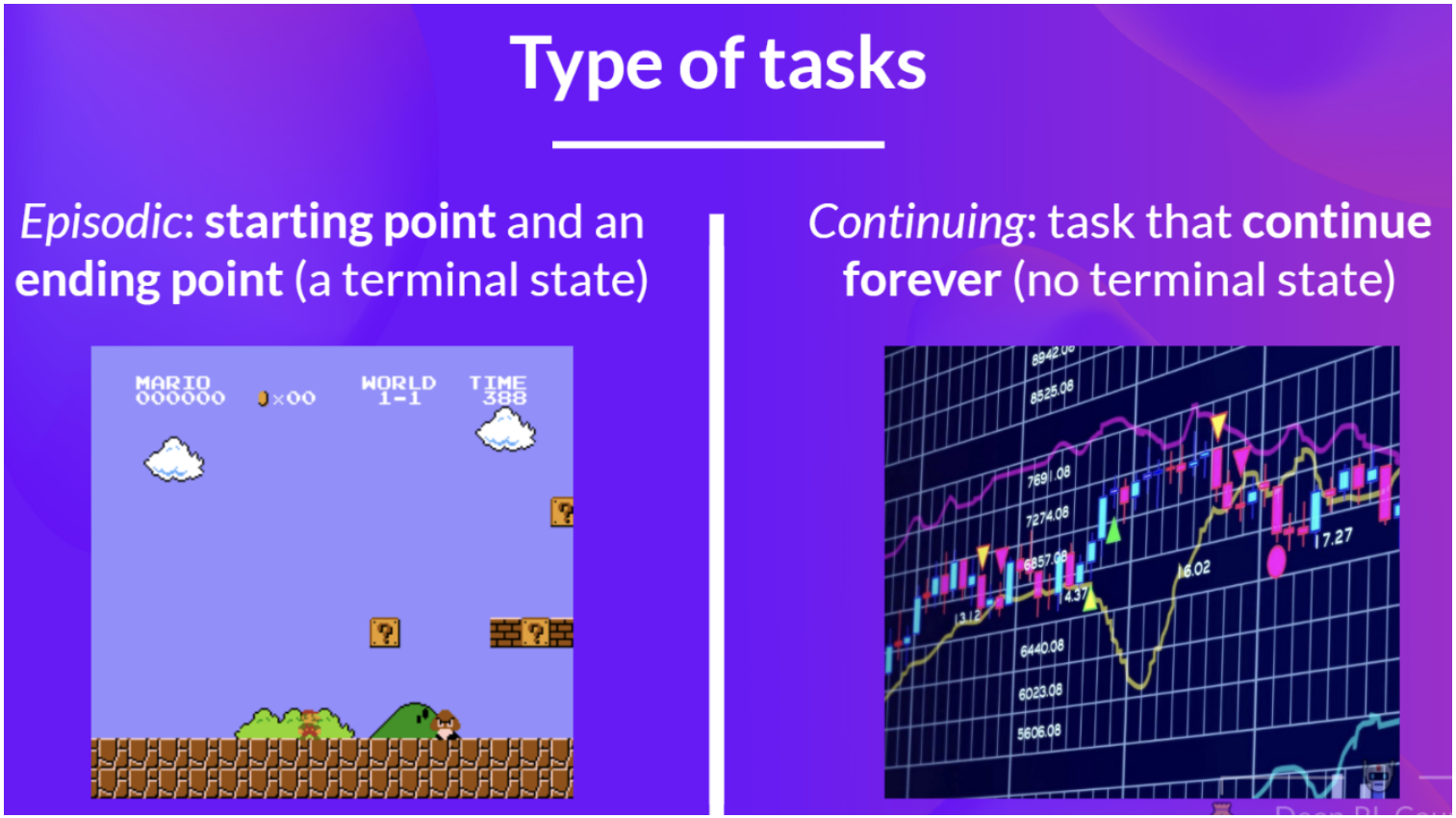


Figure: A task is an instance of a Reinforcement Learning problem.

The Exploration/Exploitation trade-off

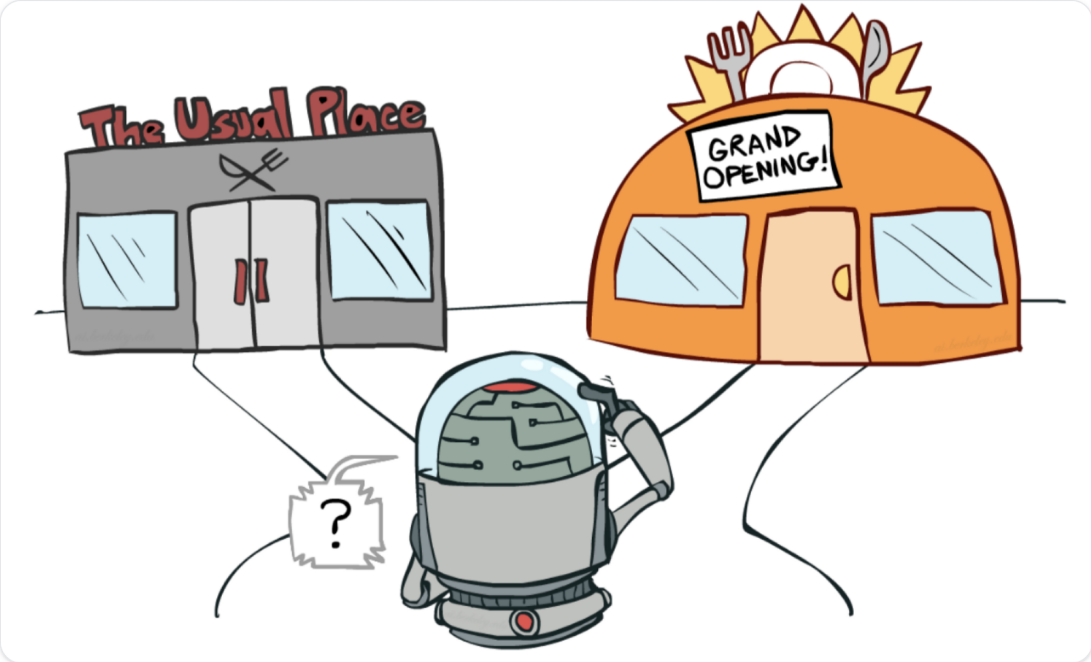


Figure: Think of a real problem: the choice of picking a restaurant

The Exploration/Exploitation trade-off

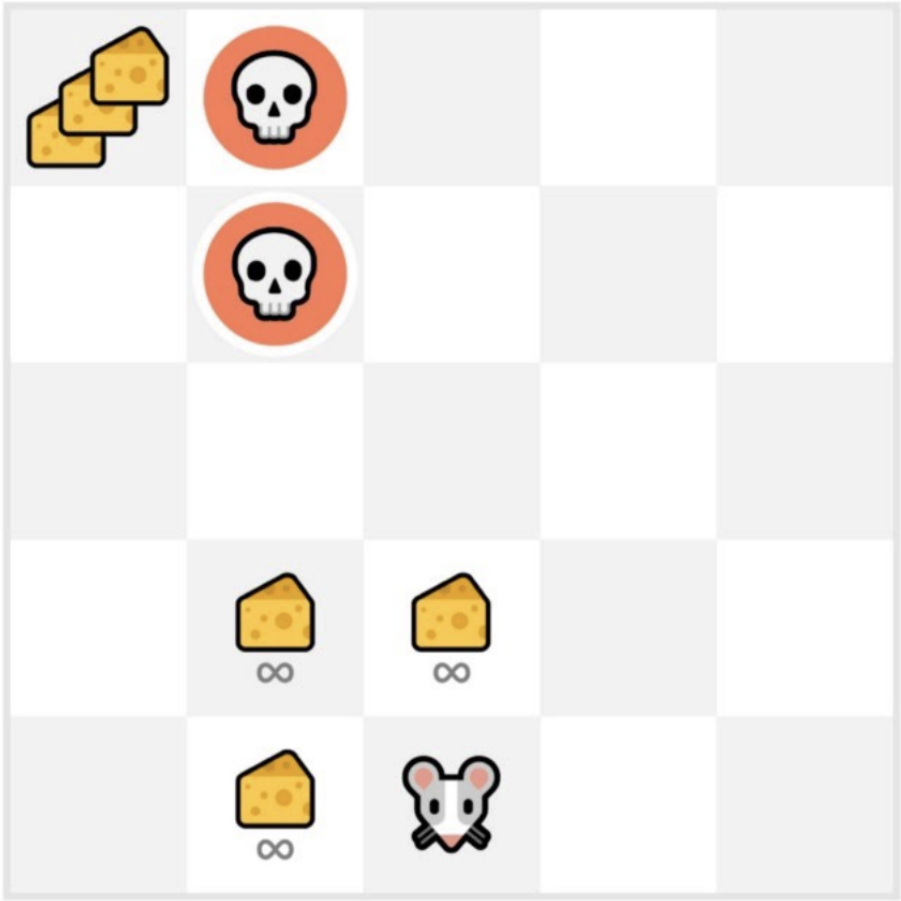


Figure: We need to balance how much we *explore* the environment and how much we *exploit* what we know about the environment.

The Exploration/Exploitation trade-off

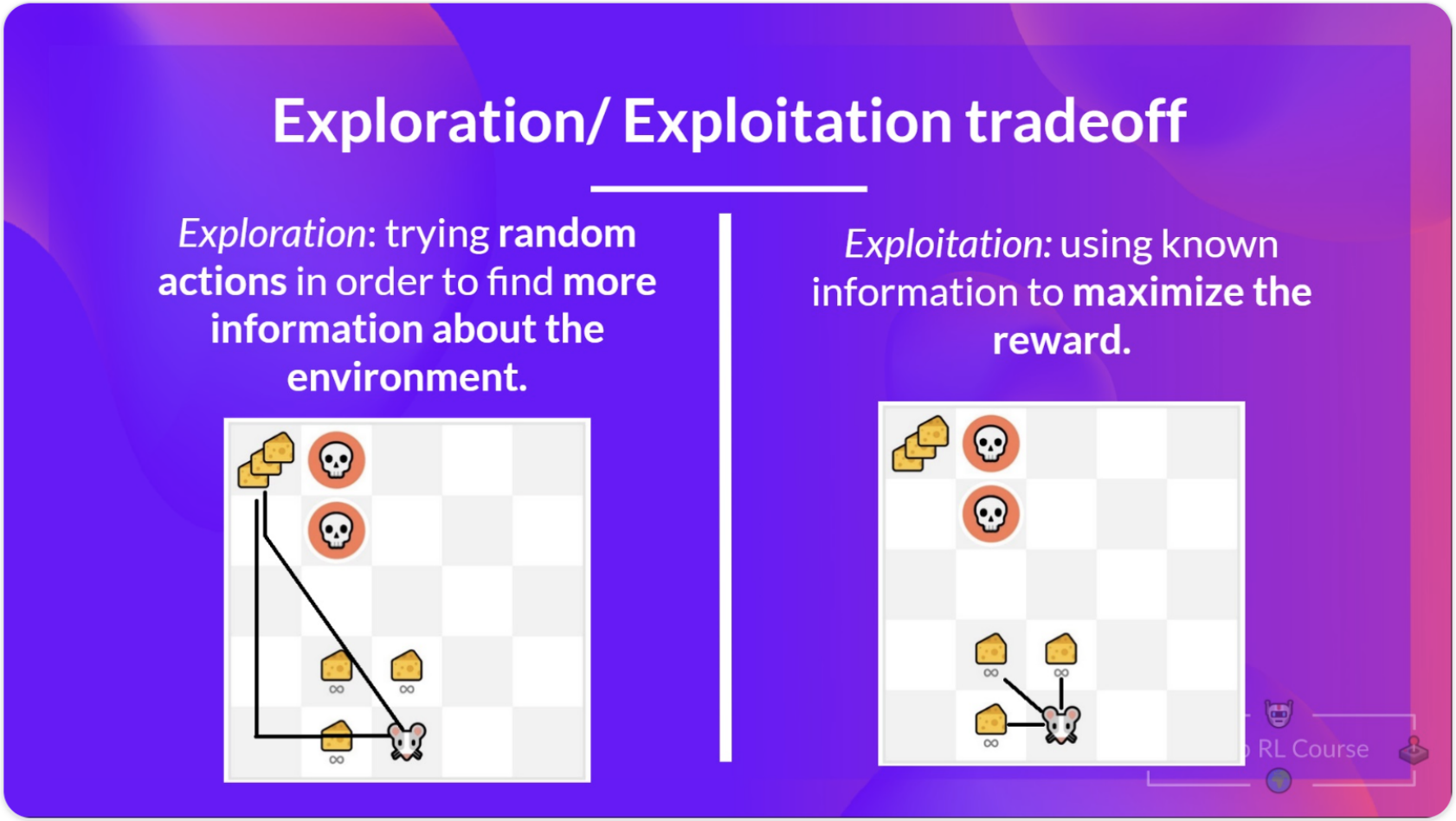


Figure: Exploration/Exploitation trade-off

Approaches for solving RL problems

- How do we build an RL agent that can select the actions that maximize its expected cumulative reward?
- The Policy π is the brain of our Agent, it's the function that tells us what action to take given the state we are in.
- The policy π defines the agent's behavior at a given time.

Objective

This Policy π is the function we want to learn, our goal is to find the optimal policy $\pi*$, the policy that maximizes expected return when the agent acts according to it. We find this $\pi*$ through **training**.

Approaches for solving RL problems

There are two approaches to train our agent to find this optimal policy $\pi*$:

- Directly, by teaching the agent to learn which action to take, given the current state: **Policy-Based Methods**.
- Indirectly, teach the agent to learn which state is more valuable and then take the action that leads to the more valuable states: **Value-Based Methods**.

Policy-Based Methods

In Policy-Based methods, we learn a policy function directly.
We have two types of policies:

- Deterministic: a policy at a given state will always return the same action.

$$a = \pi(s)$$

- Stochastic: outputs a probability distribution over actions.

$$\pi(a|s) = P[A|s]$$

Policy-Based Methods

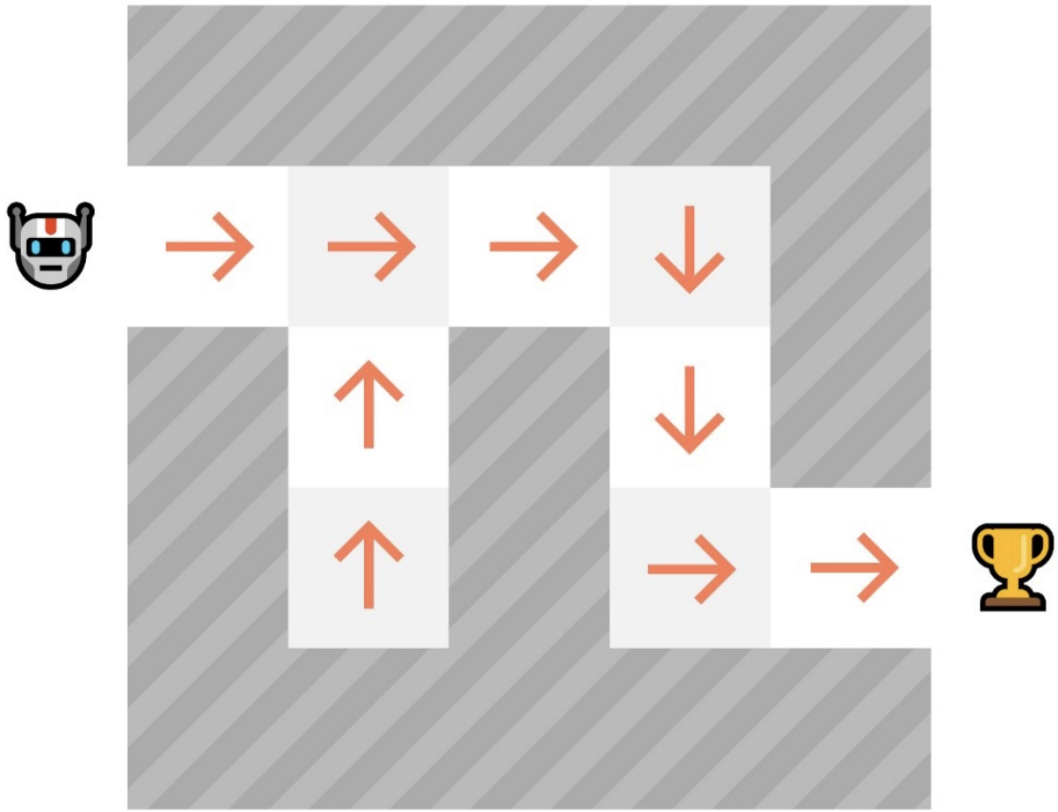


Figure: the policy (deterministic) directly indicates the action to take for each step.

Value-based methods

- In value-based methods, instead of learning a policy function, we learn a value function that maps a state to the expected value of being at that state.
- The value of a state is the expected discounted return the agent can get if it starts in that state, and then acts according to our policy.
- "Act according to our policy" just means that our policy is "**going to the state with the highest value**".

Value-based methods

$$\underbrace{v_\pi(s)}_{\text{Value function}} = \underbrace{\mathbb{E}_\pi\left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots\right]}_{\text{Expected discounted return}} \mid \underbrace{S_t = s}_{\text{Starting at state s}}$$

Value-based method

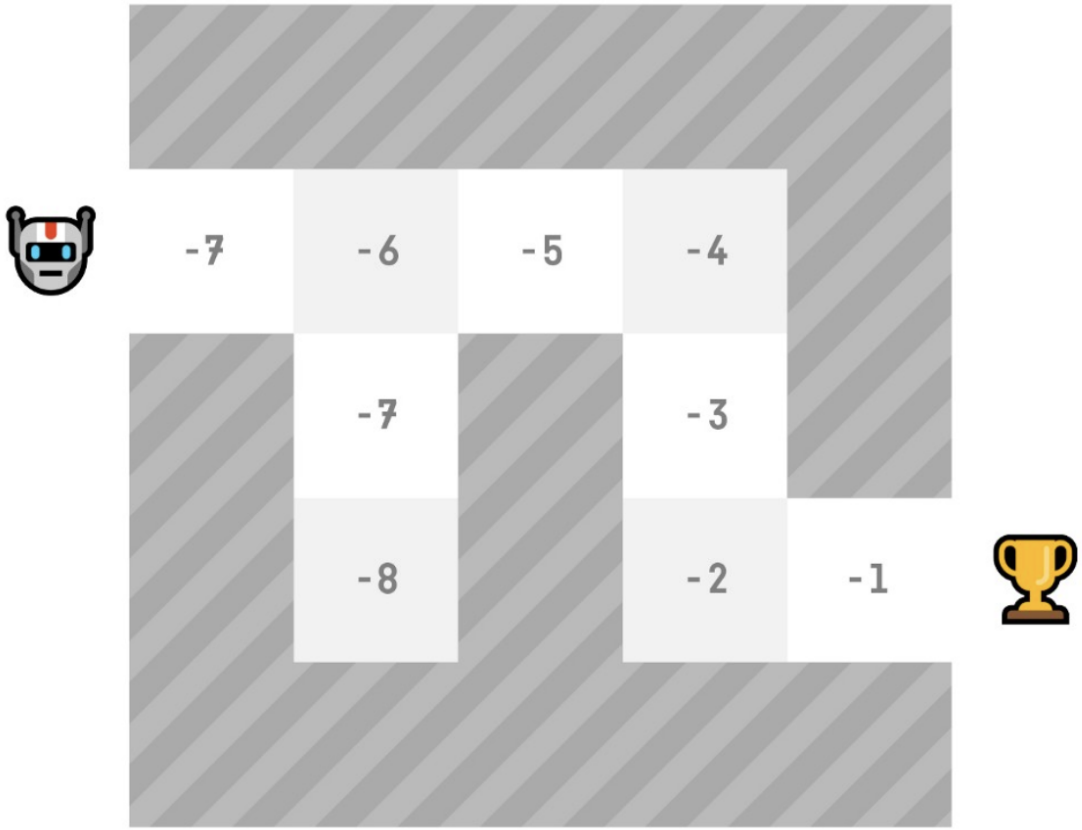


Figure: At each step our policy will select the state with the biggest value defined by the value function: -7, then -6, then -5 (and so on) to attain the goal.

Two main types of RL methods

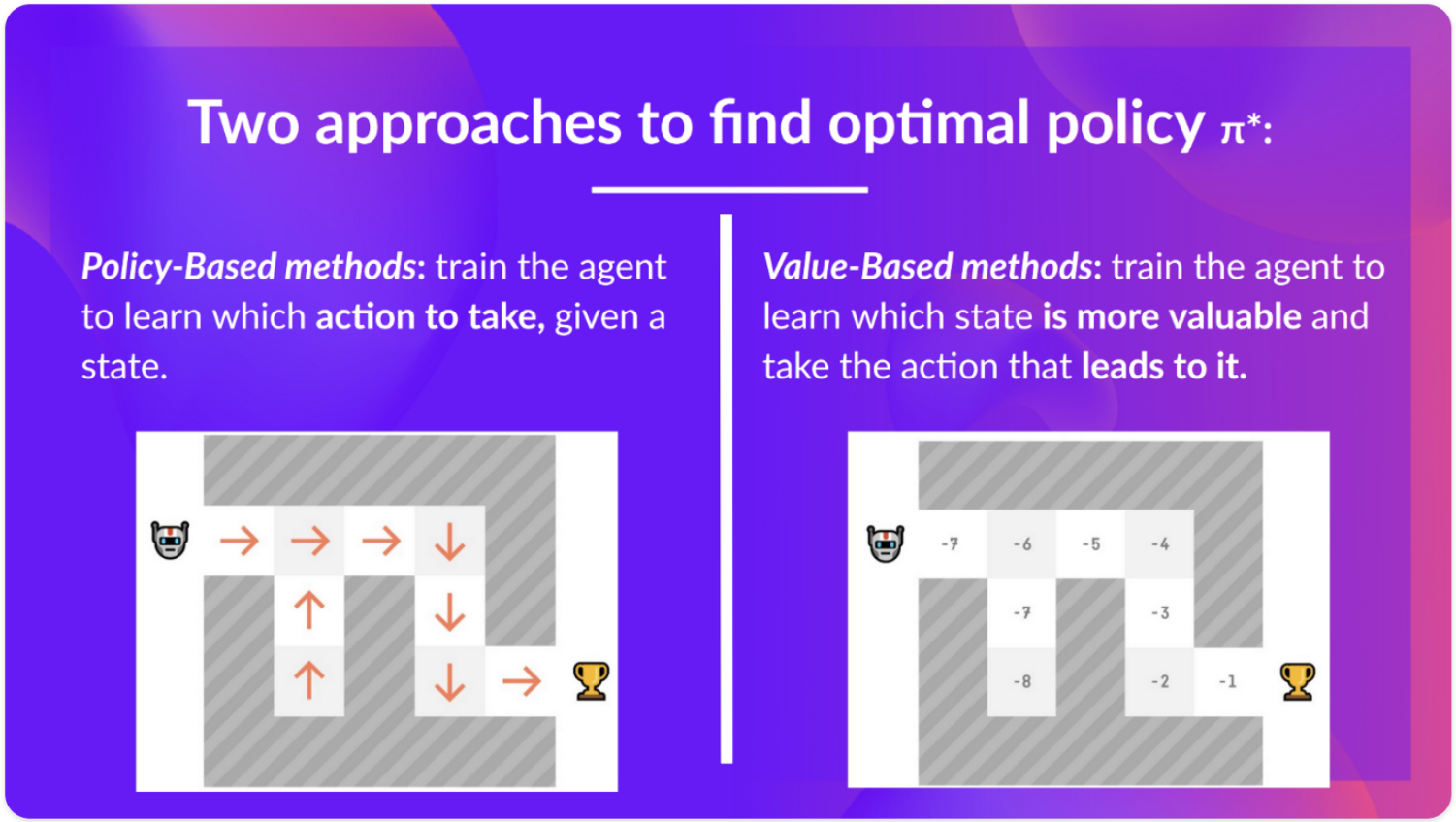


Figure: **Policy-based methods**: Train the policy directly to learn which action to take given a state.
Value-based methods: Train a value function to learn which state is more valuable and use this value function to take the action that leads to it.

value-based methods

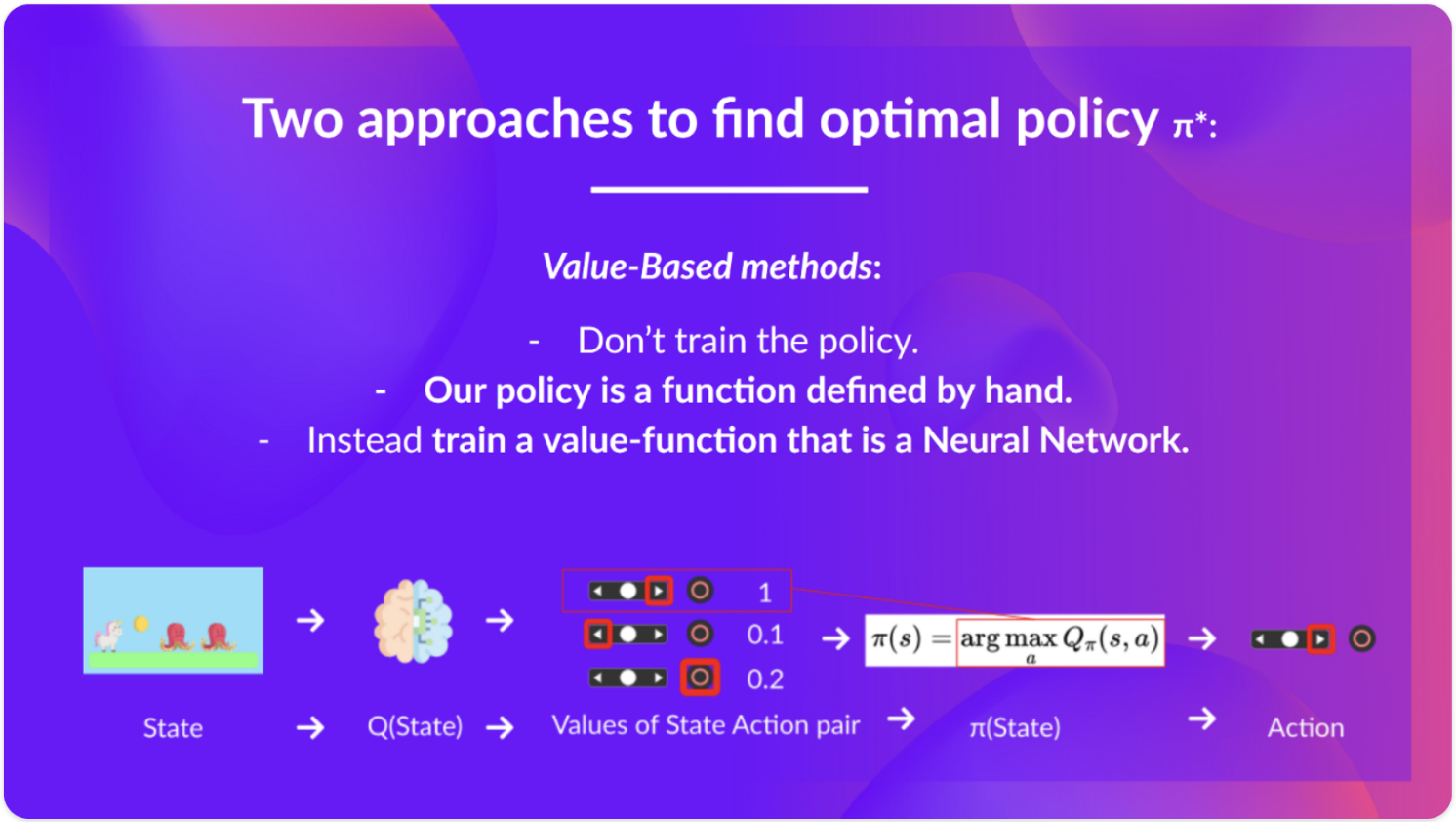
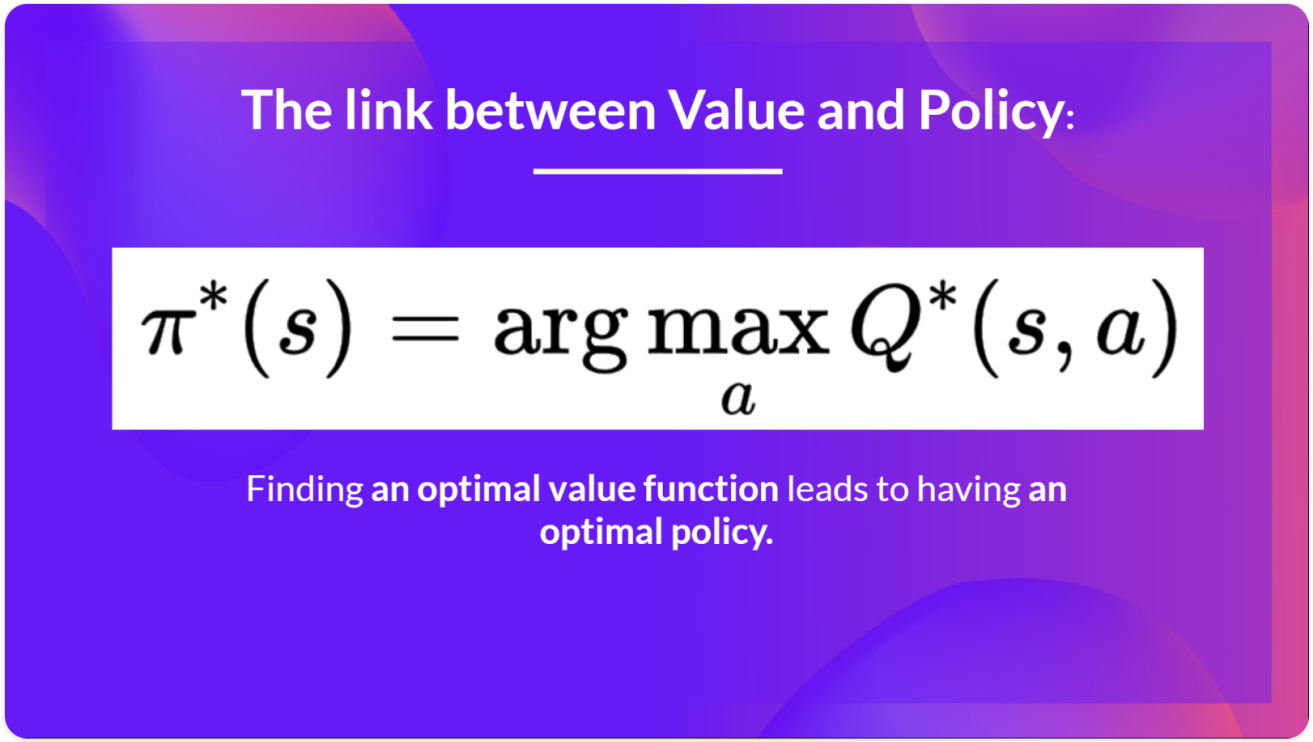


Figure: Given a state, our action-value function (that we train) outputs the value of each action at that state. Then, our pre-defined Greedy Policy selects the action that will yield the highest value given a state or a state action pair.

value-based methods

- In policy-based training, the optimal policy (denoted $\pi*$) is found by training the policy directly.
- In value-based training, finding an optimal value function (denoted $Q*$ or $V*$) leads to having an optimal policy.



Types of value-based functions:

- The state-value function
- The action-value function

Math behind Q-learning

$$V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$$

New value of state t

Former estimation of value of state t

Learning Rate

Reward

Discounted value of next state

TD Target

Therefore, our $Q(S_t, A_t)$ **update formula goes like this:**

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[R_{t+1} + \gamma max_a Q(S_{t+1}, a) - Q(S_t, A_t)]$$

New Q-value estimation

Former Q-value estimation

Learning Rate

Immediate Reward

Discounted Estimate optimal Q-value of next state

Former Q-value estimation

TD Target

TD Error